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SUB: OSSP(Practical)

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsbk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Output:

```
vaibhavi@VYSRI:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

vaibhavi@VYSRI:~$ nano fork_exec.c
vaibhavi@VYSRI:~$ gcc fork_exec.c -o fork_exec
vaibhavi@VYSRI:~$ ./fork_exec
Enter a Linux command: date

Parent Process
Parent PID: 641
Child PID : 642

Child Process
Child PID : 642
Parent PID: 641
Fri Aug 7 09:22:41 UTC 2026

Child process completed.
vaibhavi@VYSRI:~$ uname -a
Linux VYSRI 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
vaibhavi@VYSRI:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1335U
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4991.99
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
```

```
vaibhavi@VYSRI: ~  
  
BogoMIPS: 4991.99  
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtr  
r pge mca cmov pat pse36 clflush mmx fxsr sse s  
se2 ss ht syscall nx pdpe1gb rdtscp lm constant  
_tsc rep_good nopl xtopology tsc_reliable nonst  
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s  
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p  
opcmt tsc_deadline_timer aes xsave avx f16c rdr  
and hypervisor lahf_lm abm 3dnowprefetch ssbd i  
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi  
d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi  
2 erms invpcid rdseed adx smap clflushopt clwb  
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni  
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo  
vdiri movdir64b fsrm md_clear serialize ibt flu  
sh_llld arch_capabilities  
  
Virtualization features:  
Virtualization: VT-x  
Hypervisor vendor: Microsoft  
Virtualization type: full  
Caches (sum of all):  
L1d: 288 KiB (6 instances)  
L1i: 192 KiB (6 instances)  
L2: 7.5 MiB (6 instances)  
L3: 12 MiB (1 instance)  
NUMA:  
NUMA node(s): 1  
NUMA node0 CPU(s): 0-11  
Vulnerabilities:  
Gather data sampling: Not affected  
Ghostwrite: Not affected  
Indirect target selection: Not affected  
Itlb multihit: Not affected  
L1tf: Not affected  
Mds: Not affected  
Meltdown: Not affected  
Mmio stale data: Not affected  
Old microcode: Not affected  
Reg file data sampling: Mitigation; Clear Register File  
Retbleed: Mitigation; Enhanced IBRS  
Spec rstack overflow: Not affected
```

```
vaibhavi@VYSRI: ~  
root      46      1  0 09:16 ?      00:00:00 /usr/lib/systemd/systemd  
systemd+  78      1  0 09:16 ?      00:00:00 /usr/lib/systemd/systemd  
root      84      1  0 09:16 ?      00:00:00 /usr/lib/systemd/systemd  
root     140      1  0 09:16 ?      00:00:00 /bin/sh /usr/lib/systemd  
root     144      1  0 09:16 ?      00:00:00 /usr/sbin/cron -f -P  
message+ 145      1  0 09:16 ?      00:00:00 @dbus-daemon --system --  
root     152      1  0 09:16 ?      00:00:00 /usr/bin/python3 /usr/bi  
root     169      1  0 09:16 ?      00:00:00 /usr/lib/systemd/systemd  
root     191      1  0 09:16 ?      00:00:00 /usr/libexec/wsl-pro-ser  
syslog   223      1  0 09:16 ?      00:00:00 /usr/sbin/rsyslogd -n -i  
_chrony  261     140  0 09:16 ?      00:00:00 /usr/sbin/chronyd -n -F  
_chrony  263     261  0 09:16 ?      00:00:00 /usr/sbin/chronyd -n -F  
root     270      1  0 09:16 ?      00:00:00 /usr/bin/python3 /usr/sh  
root     275      1  0 09:16 tty1    00:00:00 /usr/sbin/agetty --nores  
root     319      2  0 09:16 ?      00:00:00 login -- vaibhavi  
vaibhavi 362      1  0 09:16 ?      00:00:00 /usr/lib/systemd/systemd  
vaibhavi 364     362  0 09:16 ?      00:00:00 (sd-pam)  
vaibhavi 391     319  0 09:16 pts/1    00:00:00 -bash  
root     529      2  0 09:21 ?      00:00:00 /init  
root     530     529  0 09:21 ?      00:00:00 /init  
vaibhavi 535     530  0 09:21 pts/0    00:00:00 -bash  
vaibhavi 694     535  0 09:24 pts/0    00:00:00 ps -ef  
vaibhavi@VYSRI:~$ top  
top - 09:26:39 up 10 min, 1 user, load average: 0.01, 0.02, 0.00  
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie  
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.  
MiB Mem : 7762.6 total, 7224.8 free, 527.9 used, 161.2 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7234.8 avail Mem  


| PID | USER     | PR | NI | VIRT  | RES   | SHR   | S | %CPU | %MEM | TIME+   |
|-----|----------|----|----|-------|-------|-------|---|------|------|---------|
| 1   | root     | 20 | 0  | 24164 | 15340 | 11576 | S | 0.0  | 0.2  | 0:00.52 |
| 2   | root     | 20 | 0  | 3180  | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.00 |
| 6   | root     | 20 | 0  | 3180  | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 |
| 46  | root     | 19 | -1 | 50428 | 16692 | 15408 | S | 0.0  | 0.2  | 0:00.13 |
| 78  | systemd+ | 20 | 0  | 22416 | 14496 | 12000 | S | 0.0  | 0.2  | 0:00.07 |
| 84  | root     | 20 | 0  | 35364 | 12308 | 9228  | S | 0.0  | 0.2  | 0:00.36 |
| 140 | root     | 20 | 0  | 2888  | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.03 |
| 144 | root     | 20 | 0  | 4472  | 3120  | 2876  | S | 0.0  | 0.0  | 0:00.00 |
| 145 | message+ | 20 | 0  | 8796  | 5400  | 4720  | S | 0.0  | 0.1  | 0:00.03 |
| 152 | root     | 20 | 0  | 44096 | 29716 | 14576 | S | 0.0  | 0.4  | 0:00.19 |
| 169 | root     | 20 | 0  | 18436 | 9392  | 8148  | S | 0.0  | 0.1  | 0:00.07 |


```

```

MiB Mem : 7762.6 total, 7224.8 free, 527.9 used, 161.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7234.8 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	24164	15340	11576	S	0.0	0.2	0:00.52
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.00
6	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00
46	root	19	-1	50428	16692	15408	S	0.0	0.2	0:00.13
78	systemd+	20	0	22416	14496	12000	S	0.0	0.2	0:00.07
84	root	20	0	35364	12308	9228	S	0.0	0.2	0:00.36
140	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.03
144	root	20	0	4472	3120	2876	S	0.0	0.0	0:00.00
145	message+	20	0	8796	5400	4720	S	0.0	0.1	0:00.03
152	root	20	0	44096	29716	14576	S	0.0	0.4	0:00.19
169	root	20	0	18436	9392	8148	S	0.0	0.1	0:00.07
191	root	20	0	1857908	15084	12552	S	0.0	0.2	0:00.11
223	syslog	20	0	220548	5848	4564	S	0.0	0.1	0:00.06
261	_chrony	20	0	23592	11024	7100	S	0.0	0.1	0:00.04
263	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.00
270	root	20	0	123460	32640	18032	S	0.0	0.4	0:00.13
275	root	20	0	5492	2732	2548	S	0.0	0.0	0:00.00
319	root	20	0	8648	5568	4724	S	0.0	0.1	0:00.01
362	vaibhavi	20	0	22340	13528	10912	S	0.0	0.2	0:00.07
364	vaibhavi	20	0	22912	4096	2096	S	0.0	0.1	0:00.00
391	vaibhavi	20	0	6136	5420	3864	S	0.0	0.1	0:00.01
529	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
530	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.01
535	vaibhavi	20	0	6268	5536	3864	S	0.0	0.1	0:00.04
704	vaibhavi	20	0	8172	6000	3888	R	0.0	0.1	0:00.06

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

OUTPUT:

```

vaibhavi@VYSRI:~$ nano filecopy.c
vaibhavi@VYSRI:~$ nano sample.txt
vaibhavi@VYSRI:~$ gcc filecopy.c -o filecopy
vaibhavi@VYSRI:~$ ./filecopy
File copied successfully.
vaibhavi@VYSRI:~$ cat copy.txt
Hello Ubuntu
Operating Systems Lab
Using open(), read(), write(), and close()
vaibhavi@VYSRI:~$ strace -V
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx32-mpers
vaibhavi@VYSRI:~$ cat sample.txt
Hello Ubuntu
Operating Systems Lab
Using open(), read(), write(), and close()
vaibhavi@VYSRI:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffe45276c38 /* 25 vars */) = 0
brk(NULL)                               = 0x654f7b91d000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x744d68006000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19947, ...}) = 0

```

```
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x744d68006000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19947, ...}) = 0
mmap(NULL, 19947, PROT_READ, MAP_PRIVATE, 3, 0) = 0x744d68001000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x744d67fce000
mmap(0x744d67fd5000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x744d67fd5000
mmap(0x744d67ff6000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x744d67ff6000
mmap(0x744d67ffd000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x744d67ffd000
mmap(0x744d67fff000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x744d67fff000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x744d67fa0000
mmap(0x744d67fa4000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x744d67fa4000
mmap(0x744d67fc8000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x744d67fc8000
mmap(0x744d67fcc000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x744d67fcc000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x744d67e7a000
mmap(0x744d67e8c000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x744d67e8c000
mmap(0x744d67f2c000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x744d67f2c000
mmap(0x744d67f9e000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x744d67f9e000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x744d67c00000
mmap(0x744d67c28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x744d67c28000
mmap(0x744d67dc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x744d67dc0000
mmap(0x744d67e0e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x744d67e0e000
mmap(0x744d67e14000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x744d67e14000
```